hiperspektral görüntü işleme

220 dalga boyu ner kanal

verinin boyutunu azaltmak için

① Feature selection

$\{x^t, r^t\}_{t=1}^N$

$x_{N\times 1}^t \rightarrow x_{N_2\times 1}^t \quad N_2 < N_1$

Subset Selection

$F = \emptyset$

while not converge:

for $x_i \notin F$

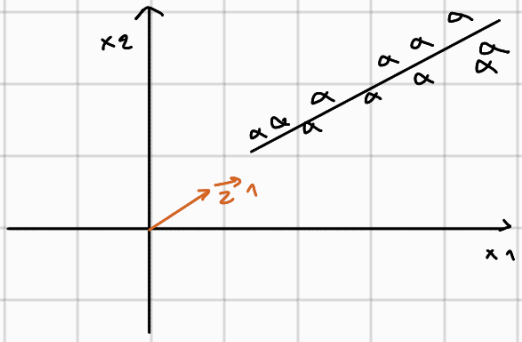
$F_i \leftarrow F \cup x_i$

$d_i(x \mid F_i)$

$i = \text{argmin}\ d_i(x_i \mid F_i)$

$F \leftarrow F_i$

Başka bir yöntem var O da PCA



regresyonda $\sum_{t=1}^N (\hat{r}^t - r^t)^2$

$(X_z)^T (X_z) = z^T X^T X z$

Σ

pozitif yarıtanımlı

$$z_1 = \arg\max_z \underbrace{\left(z^T \Sigma z - \alpha (z_1^T z_1 - 1) \right)}_{L(z|x)}$$

$$\frac{\partial L}{\partial z} = 0$$

① $\frac{\vec{x_i} - \vec{\mu}}{\Sigma}$ ② [min, max]

$$2\Sigma z - 2\alpha z = 0$$

$$2\Sigma z = 2\alpha z$$

$$\boxed{\Sigma z = \alpha z}$$

$$z^T \Sigma z = z^T \alpha z$$

$$= \alpha z^T \cdot z = \alpha$$

$$z_1 \; \Sigma z_1 = \alpha_1^2 \qquad \boxed{\lambda_1}$$

$$z_2 \; \Sigma z_2 = \alpha_2^2 \qquad \boxed{\lambda_2}$$

$x_{N \times 1}$

$$z_2 = \arg\max_z \left(z^T \Sigma z - \alpha (z^T z - 1) - \beta (z^T \cdot z_1) \right)$$

$z_1^T /$ $\quad 2\Sigma z - 2\alpha z - \beta z_1 = 0$

$$2 z_1^T \Sigma z - 2\alpha z_1^T z - \beta z_1^T z_1 = 0$$

$$2 z_1^T \Sigma z - 2\alpha z_1^T \cdot z = \beta$$

$$2 z_1^T \Sigma z - 2\alpha z_1^T z = \beta$$

$$\underline{2 z_1^T \Sigma z}$$

Principle Component analizi ← Dimension reduction

$N \rightarrow M$

$z_1, z_2, z_3, z_4, \cdots, z_M$

$X \cdot z_1 = \begin{bmatrix} y_1 \\ \\ \end{bmatrix}$

$X \cdot z_2 = \begin{bmatrix} y_2 \\ \\ \end{bmatrix}$

$X \cdot z_m = \begin{bmatrix} y_m \\ \\ \end{bmatrix}$

$\underbrace{[x_1\ x_2\ x_3 \cdots]}_{X^T} \quad [z_1\ z_2\ z_3 \cdots z_m]$

$X \cdot shape = (100, 200, 224)$

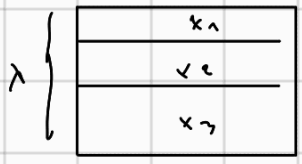
$X.view(-1, 224)$

kullanın

Haftaya perceptron bakacağız

Feature Embedding

$WDW^T = X \cdot X^T$



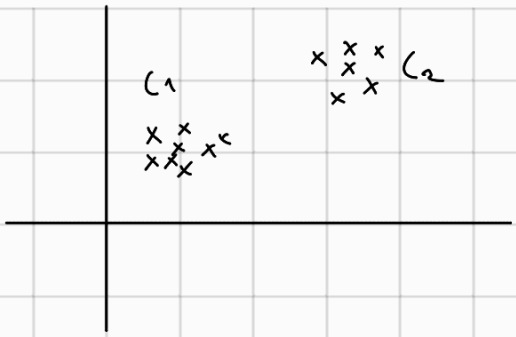
Eigen faces

Singular value composition

SVD: $x = [x_1\ x_2 \cdots x_N]_{N \times d}$

$= V A W^T$

$N \times f \quad f \times f \quad f \times d$



→ Linear Discriminant Analysis

$z = w^T x$

$$m_1 = \frac{\sum_t w^T x^t r^t}{\sum_t r^t} = w^T \cdot \vec{m_1}$$

$$m_2 = \frac{\sum_t w^T r^t (1-r^t)}{\sum_t (1-r^t)} = w^T \vec{m_2}$$

$$s_1^2 = \sum_t (w^T x^t - m_1)^2 r^t$$

$$s_2^2 = \sum_t (w^T x^t - m_2)^2 (1-r^t)$$

$$J(w) = \frac{(m_1 - m_2)^2}{s_1^2 + s_2^2} = \frac{(w^T \vec{m_1} - w^T \cdot \vec{m_2})}{s_1^2 + s_2^2}$$

$$= \frac{w^T (\vec{m_1} - \vec{m_2})^T (\vec{m_1} - \vec{m_2}) w}{s_1^2 + s_2^2}$$

$$= \frac{w^T S_B w}{s_1^2 + s_2^2}$$

$$s_1^2 = \sum_t (w^T x^t - m_1)^2 r^t$$

$$= w^T \cdot S_1 \cdot w \qquad S_1 = \sum_t w^t (x^t - \vec{m_1})(x^t - \vec{m_1})^T r^t$$

$$s_1^2 + s_2^2 = w^T S_w w = \vec{w} (S_1^2 + S_2^2) w$$

$$J(w) = \frac{w^T S_B \cdot w}{w^T \cdot S_w w}$$

$$\frac{\partial S(w)}{\partial w} = 0$$

$$= \frac{w^T (\vec{m_1} - \vec{m_2})}{w^T S_w w} 2$$

Soft kmeans

$P(x|C_i) = ??$ soft label

bellibir eşiğin altında kullanıyoruz

Expectation maximization



hierarchical clustering

DB scan

